

OM NATHWANI

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Education

University of Waterloo & Wilfrid Laurier University
BCS & BBA, Computer Science and Business Administration (Co-op)

Waterloo, Ontario
2024 - 2029

- **GPA:** 3.7/4.0
- **Extra-curricular:** Dual-Varsity Athlete in Cross Country and Track + LazSoc Software Developer
- **Awards/Scholarships:**
 - President's Gold Scholarship (95-100% entry average) - *\$4,000 per year*
 - **BDO Future Leaders Challenge Finalist** ↗

Technical Skills

- **Languages:** Java, Python, Racket, JavaScript, TypeScript, PostgreSQL, HTML, Lua, CSS, C, C++, TailwindCSS
- **Libraries/Frameworks:** Flask, FastAPI, GraphQL, SQLAlchemy, Pytorch, TensorFlow, Pandas, NumPy, Langchain, ChromaDB, SciKitLearn, Express.js, Node.js, Next.js, React.js
- **Developer Tools:** Vercel, Docker, Git, Github Actions/Cron, GCP, AWS, Supabase

Experience

Software Developer

Jan 2026 – Apr 2026

Dawson Partners | Python, FastAPI, SQLAlchemy, GCP, Apache Airflow, LangSmith, Pandas, NumPy, Kyriba, GraphQL, Git

Toronto, Ontario

- Increased **MCP system correctness** to **98%** by building **validation and error-handling workflows** for our financial models
- Improved **agent reasoning** by evaluating *LangSmith* traces and documenting **financial model** business concepts, inputs, and outputs
- Standardized **tax doc extraction** across Dawson's firmwide data pipeline by integrating nuanced workflows into **inter, staging, and curated layers**
- Enabled final filing submission, and **tax-form scalability** by designing page-level *SQLAlchemy* models and *Apache Airflow* DAGs for **270+** extracted fields
- Protected **200+ weekly CF payment submissions** by extending **third-party Kyriba integration** with rebilling payment support, webhook-based status tracking, and invoice submission refactors
- Supported **firmwide KT** by researching *GraphQL* mutations to build CRUD architecture for a **dynamic** internal insights page

Software Engineer Intern

May 2025 – August 2025

Covalense Digital | Python, FastAPI, CLIP, pdfplumber, Pytesseract, PostgreSQL, Psychopg2, ChromaDB, Ollama, OCR, Git

Mississauga, Ontario

- Developed and demoed a **secure, RBAC-enabled internal document chatbot** to **senior executives** that lets employees query, search, and interact with company data in natural language while respecting access permissions
- Wired internal tools with *FastAPI* and **MCP queues** by building a secure ingestion and search backend, which allowed employees to query indexed content through **RBAC filtering** and handled over **500** documents and **1,000+** images
- Integrated *CLIP* and base64 encoding with vector search by adding **image embedding and retrieval**, which enabled users to surface exact image chunks
- Prioritized performance and safety by researching **50+ libraries** to select **MIT-compliant** tools like *pdfplumber* and *Chroma*

Software Engineer

Sept. 2024 – May 2025

Wat.AI | Python, AWS, RESTful APIs, OpenAI, RAG

Waterloo, Ontario

- Defined **ML pipeline structure** by researching Retrieval-Augmented Generation and Agentic AI using **30+** papers and UWaterloo databases, which informed backend decisions
- Optimized prompt handling and response logic by caching inputs and improving **API** efficiency, which cut latency by over **50%**

Projects

Bollyguess ↗ | Next.js, Node.js, TailwindCSS, React, Python, PostgreSQL, Supabase, Docker, GitHub Actions, Vercel **July 2025**

- Launched a daily Bollywood song guessing game by building and promoting the app through social channels, hitting over **200+** plays in the first week
- Wrote data cleaning and seeding scripts in *Node.js*, to scale *PostgreSQL* database with **2,400+** songs and **7,500+** artists.
- Displayed **daily leaderboards** by tracking scores in *Supabase* and connecting it through *Next.js* API routes, to encourage **competition**
- Set up **automated daily updates**, eliminating manual work post-launch, using *GitHub Actions* and *cron* jobs
- Randomized daily games using *Python* and many-to-many logic in *PostgreSQL*, which delivered **fresh gameplay every day**

Audio Segment Classifier ↗ | Python, PyTorch, TensorFlow, Librosa, FFmpeg, Pandas, SciKitLearn

July 2025

- Trained a **4-layer feedforward neural network** with a final sigmoid output using *PyTorch* to **dynamically** label clips as good (1) or bad (0), automating ideal segment selection
- Extracted audio features using *Librosa* by building a preprocessing pipeline, which gave the model reliable input vectors
- Validated performance with an **85% F1-score** and **confusion matrix** by testing against labeled data, confirming readiness
- Ran the classifier on **2,400+** segments by **integrating it with Bollyguess**, improving clip accuracy by over **90%**